

HARRISBURG-RALEIGH, IL, USA

WMO: 744652

Lat: 37.811N

Lon: 88.549W

Elev: 121

StdP: 99.88

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.7	-17.7	0.8	-10.7	-15.1	1.0	-8.2	11.6	8.2	10.5	6.9	2.2	350	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.1	24.3	32.8	24.0	32.1	23.8	26.7	31.2	25.8	30.3	25.1	29.5	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.7	29.1	24.7	20.0	28.4	23.8	18.9	27.3	83.9	31.0	80.1	30.4	76.9	29.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.4	8.3	7.3	-17.0	36.1	3.1	1.9	-19.2	37.5	-21.1	38.7	-22.8	39.7	-25.0	41.1	(4)
(5)				-17.5	27.5	2.9	1.1	-19.6	28.3	-21.3	29.0	-22.9	29.7	-25.0	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	0.9	2.4	8.4	14.3	19.4	23.9	25.1	24.5	21.0	14.5	7.9	3.5	(6)
(7)		DBStd	9.86	6.19	6.14	5.97	4.99	4.39	2.97	2.76	2.77	3.74	5.09	5.27	5.13	(7)
(8)		HDD10.0	959	289	223	103	18	1	0	0	0	14	100	212		(8)
(9)		HDD18.3	2417	541	446	311	137	43	2	0	1	17	143	314	462	(9)
(10)		CDD10.0	2374	6	11	54	146	292	417	468	449	332	153	37	8	(10)
(11)		CDD18.3	789	0	0	3	16	76	169	211	192	98	23	1	0	(11)
(12)		CDH23.3	7664	0	0	25	148	659	1663	2094	1838	989	237	10	0	(12)
(13)		CDH26.7	2980	0	0	2	17	182	659	895	774	388	63	1	0	(13)
(14)	Wind	WSAvg	2.7	3.5	3.4	3.6	3.6	2.7	2.2	1.7	1.6	1.8	2.4	2.9	3.1	(14)
(15)	Precipitation	PrecAvg	1207	81	76	120	118	130	111	103	84	79	78	108	105	(15)
(16)		PrecMax	1677	296	170	375	332	363	323	234	243	190	263	233	333	(16)
(17)		PrecMin	808	0	17	38	44	29	12	8	9	2	1	11	14	(17)
(18)		PrecStd	197	53	39	67	60	73	64	55	55	47	53	56	61	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	21.1	26.3	28.0	32.1	35.2	37.1	36.2	34.5	31.2	24.9	19.9	(19)
(20)		MCWB	15.8	16.9	18.1	18.7	22.3	23.9	24.5	23.1	22.5	21.5	16.6	16.9		(20)
(21)		2%	DB	16.4	18.2	22.8	27.0	30.1	32.8	33.8	34.1	32.7	28.0	22.0	16.8	(21)
(22)		MCWB	13.6	14.4	16.5	18.0	21.9	23.9	25.9	23.7	22.4	20.2	16.6	14.2		(22)
(23)		5%	DB	13.8	15.3	21.1	25.1	28.8	32.1	32.5	32.5	31.2	26.1	18.9	14.1	(23)
(24)		MCWB	11.4	11.6	15.6	17.0	21.2	23.5	25.1	23.8	22.3	18.8	14.7	11.8		(24)
(25)		10%	DB	10.9	12.4	17.9	22.9	27.4	30.7	31.3	31.1	28.8	23.3	17.0	12.1	(25)
(26)		MCWB	8.4	9.3	13.7	16.3	20.3	23.3	24.4	23.6	21.1	17.5	13.0	9.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	17.8	19.2	21.1	24.7	26.8	27.9	27.3	25.3	23.8	19.7	17.7	(27)
(28)		MADB	17.7	20.3	24.1	25.9	28.5	31.7	32.8	31.7	30.5	28.2	22.5	19.6		(28)
(29)		2%	WB	14.9	15.6	17.8	19.8	23.5	25.7	27.0	26.2	24.4	22.3	17.6	15.0	(29)
(30)		MADB	16.6	17.6	21.7	24.4	27.7	30.5	31.5	30.7	29.8	26.0	19.7	16.3		(30)
(31)		5%	WB	12.0	12.6	16.3	18.8	22.4	24.9	26.2	25.3	23.5	20.5	15.6	12.3	(31)
(32)		MADB	13.2	15.0	20.2	23.2	26.8	29.6	30.6	29.8	28.4	24.3	18.3	13.8		(32)
(33)		10%	WB	8.1	9.3	14.6	17.5	21.5	24.1	25.4	24.5	22.5	18.9	13.3	9.7	(33)
(34)		MADB	10.0	12.2	17.7	20.9	25.6	28.5	29.6	28.8	26.7	22.2	16.7	11.4		(34)
(35)	Mean Daily Temperature Range	MDBR	8.6	9.0	10.4	11.6	10.8	11.0	10.8	11.5	12.7	12.3	10.6	8.4	(35)	
(36)		5% DB	MADB	12.0	13.2	13.2	13.6	12.5	12.4	12.0	13.6	15.0	14.7	13.7	11.4	(36)
(37)		MCWBR	10.1	10.2	8.0	6.9	5.4	4.8	4.9	5.1	5.9	7.1	8.9	9.5	(37)	
(38)		5% WB	MADB	10.5	12.0	12.0	10.9	10.3	10.4	10.6	11.1	12.4	12.0	11.2	10.3	(38)
(39)		MCWBR	10.0	10.5	7.9	6.6	5.3	4.9	5.1	5.1	5.6	6.5	9.2	9.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.326	0.367	0.399	0.432	0.441	0.473	0.474	0.411	0.360	0.325	0.309	(40)	
(41)		taud	2.462	2.410	2.359	2.273	2.234	2.222	2.150	2.136	2.308	2.442	2.485	2.482	(41)	
(42)		Ebn at Noon	869	903	894	881	854	843	812	801	836	853	850	849	(42)	
(43)		Edh at Noon	82	98	114	132	139	141	151	149	119	94	80	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.04	2.74	3.76	4.97	5.61	6.39	6.17	5.64	4.71	3.50	2.37	1.74	(44)	
(45)		RadStd	0.20	0.32	0.34	0.30	0.49	0.40	0.40	0.29	0.30	0.37	0.26	0.15	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77

Nomenclature: See separate page